SUPPORTING ADULT GUIDANCE

Angles on a line and at a point

Angles Tell Us Why · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Straight-Line Splitter, learners drag rays from a point on a straight line, around a point, or turn one of two crossing lines. Each angle is labelled and the total is shown: it stays 180° on a line and 360° at a point, and opposite angles share a colour and stay equal. Six “because puzzles” follow: learners find a missing angle, then choose the fact that justifies it. A stretch proves that vertically opposite angles are equal. Learners then find missing angles quickly, judge statements (including whether “it looks like 50°” is a reason), and build arrangements that meet conditions.

Driving Question: *Why must angles on a straight line make 180°?*

The mathematics, in plain terms

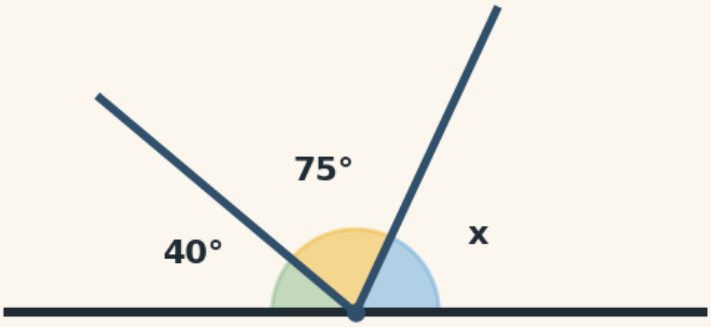
Angles on a straight line sum to 180° and angles at a point to 360°, because these are half and full turns. Vertically opposite angles are equal because each forms a straight line with the same adjacent angle (a = 180° – b = c). The lesson insists on a reason for each step; this habit is what Lesson 4 and later geometry (including circle theorems) build on. Angles with parallel lines are not part of this unit.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — On a straight line

Find x.

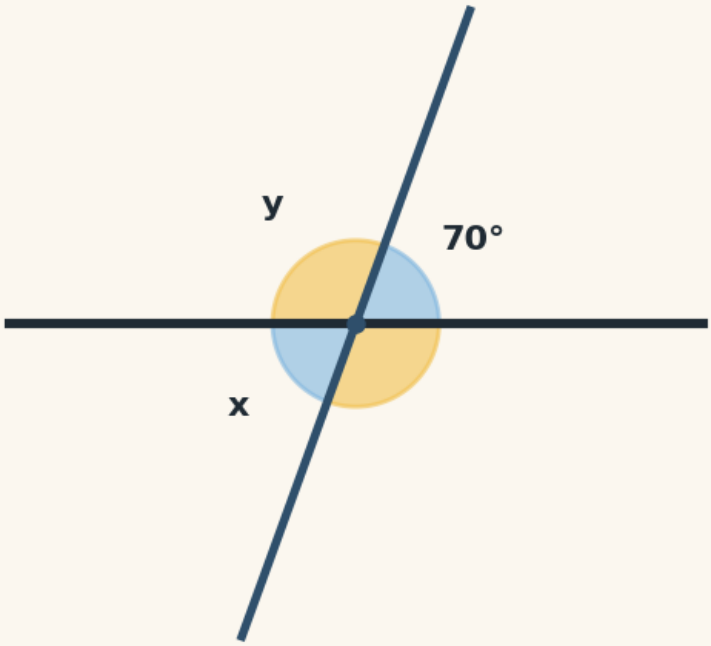


1. The three angles sit together on a straight line, so they add up to	180°
2. Add the angles you know.	75° + 40° = 115°
3. Take that from 180°.	180° – 115° = 65°

x **= 65°, because angles on a straight line add up to 180°.**

Example 2 — Where two lines cross

Find x and y.



1. x is opposite the 70° angle.	$x = 70°$
2. Reason:	vertically opposite angles are equal
3. y and 70° sit together on a straight line.	$y = 180° - 70° = 110°$
4. Reason:	angles on a straight line add up to 180°

$x = 70°$ and $y = 110°$.

See the totals stay fixed, then reason, then prove

Learners move the rays and see the totals never change before any fact is stated. The because puzzles separate finding an angle from justifying it, so the reason is never skipped. The stretch shows a proof covering every case, not just the cases tried.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	The Straight-Line Splitter and six because puzzles. Stretch: why vertically opposite angles are equal.	~16 min
Movement	Quick missing angles.	~7 min
Reflection	True or not? Including what counts as a reason.	~7 min
Creativity	Make it: arrangements that meet conditions.	~10 min
Rest	A total that never changes.	~2 min
Nutrition	Intellectual: reasoning.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using 360° on a straight line, or 180° at a point.
Ask: “Is this half a turn or a full turn?”

Giving the neighbouring angle instead of the opposite one where lines cross.
Ask: “Which angle is opposite? Which ones share a straight line?”

Giving a reason such as “it looks like 50°” or measuring a diagram marked not to scale.
Ask: “Which fact is always true here?”

Thinking all four angles are equal when two lines cross.
Ask: “Are the angles next to each other the same size? What do they add up to?”

Forgetting to add all the known angles before subtracting.
Ask: “Have you used every angle you were given?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: half and full turns (Lesson 1). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes “x = 70°” for the angle next to 70° where two lines cross. The aim is a question that separates opposite from neighbouring angles.

- “Is x opposite the 70°, or next to it?”
- “What do x and the 70° sit on together?”

- “So what must they add up to?”
- “Which angle would be 70°?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Must learners write the reason every time?
Yes, in this unit: “because …” is the habit being built. Their own words are fine, as long as the fact is right.

What about angles with parallel lines?
Those come later (Year 8 geometry, then Year 9). This lesson uses only lines, points and crossing lines.

Is the stretch required?
No. The secure core is using the three facts and naming the one used.

What comes next?
Lesson 3: angles in a triangle.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.